Assessing Fire Damage & Severity in Alberta

2024 Jasper Wildfire Complex

Sentinel-2 Imagery | Machine Learning | Deep Learning

516.7
km² burned detected

Team Members

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Why This Project Matters

35,000 ha	714,874 ha	1,230
Burned in Jasper National Park	Alberta burned in 2024	Provincial fires in 2024

1. Largest fire in Jasper National Park in over a century — destroyed nearly half the townsite
2. Satellite remote sensing provides landscape-scale burned area coverage unavailable from ground surveys
3. Accurate burned area maps are essential for post-fire ecological recovery, carbon accounting, and risk assessment
4. This project compares four ML/DL methods on the same dataset to determine which best classifies burned pixels

Sources: Parks Canada (2025); Government of Alberta (2025)

Two Complementary Data Sources

Dataset 1 — Sentinel-2 Level-2A

- 6 GeoTIFF files (pre-fire + post-fire)
- Bands: NIR (B8 10m), SWIR (B12 20m), Red (B4 10m)
- 4,515 × 4,776 pixels — 19.27M valid pixels
- CRS: EPSG:32611 (UTM Zone 11N)
- Acquisition window: before/after July 22, 2024
- Source: Microsoft Planetary Computer
- Licence: Copernicus Open Access

Dataset 2 — Alberta Wildfire CSV

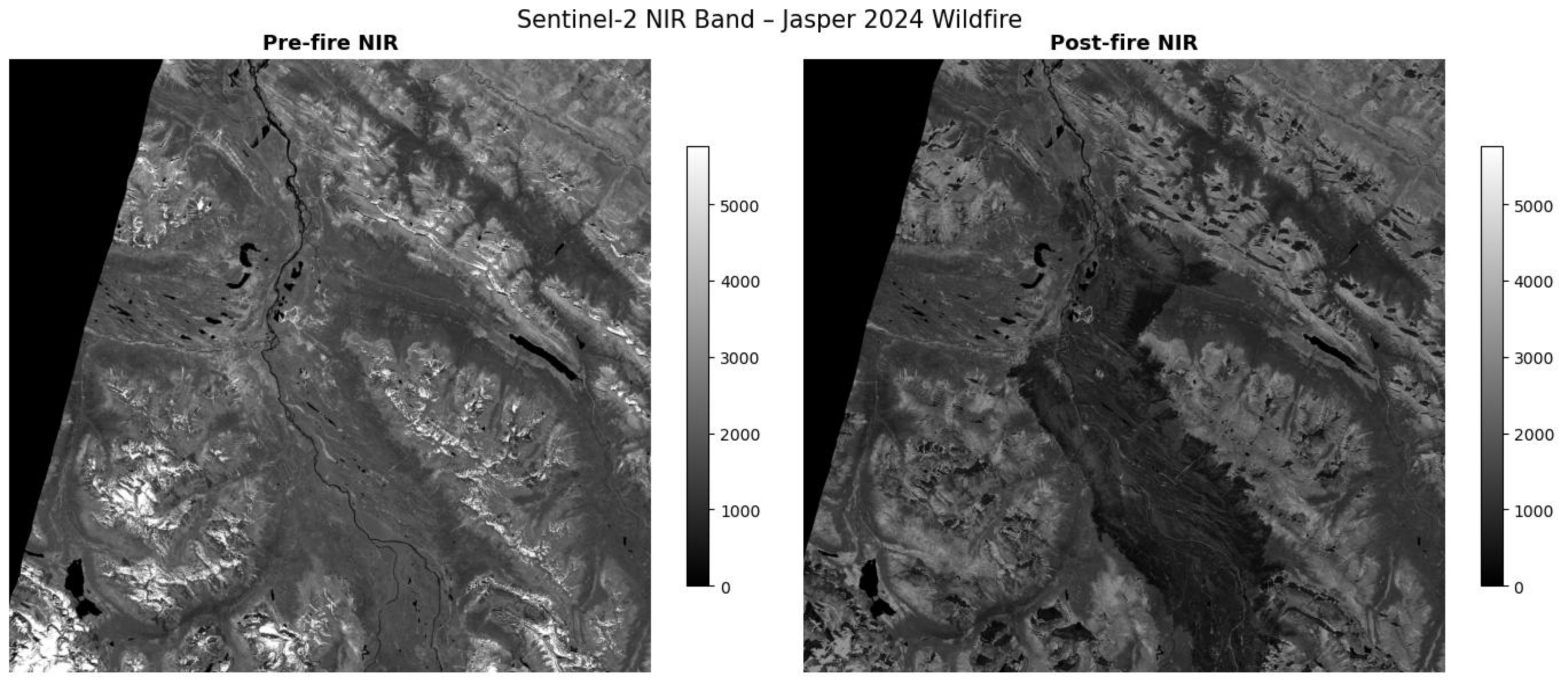
- 27,828 fire records (2006–2025), 50 columns
- Weather: temp, RH, wind speed, spread rate
- Fuel type, slope position, cause, duration
- 3 fires matched to satellite tile:

  RWF064 (6,966 ha) · RWF062 (4,051 ha)

  RWF063 (2,070 ha) — all lightning, July 22 2024
- Licence: Open Government Licence — Alberta

Government of Alberta (2025); European Space Agency (2023)

Pre- and Post-fire Images



Government of Alberta (2025); European Space Agency (2023)

METHODOLOGY

Three-Phase Analysis Pipeline

01 Phase 1

Data Preprocessing

- Load & align 6 Sentinel-2 bands
- Compute NDVI, NBR, dNBR indices
- USGS severity thresholds (4 classes)
- Build leak-free pixel dataset (19.27M rows)

02 Phase 2

Machine Learning

- Spatial column split (64/16/20)
- 8 leak-free spectral features
- Logistic Regression (baseline)
- Random Forest + MLP (pipeline)

03 Phase 3

Deep Learning (CNN)

- 9×9 pixel spatial patches (237k)
- Custom BurnCNN: 3 conv blocks
- WeightedRandomSampler for balance
- 12 epochs, Adam + StepLR

Leakage prevention: dNBR and ndvi_diff excluded from all models — both are derived directly from the burned label threshold (dNBR ≥ 0.10)

FULL PIPELINE

End-to-End Project Pipeline

Sentinel-2
6 GeoTIFFs · 10 m

Alberta Wildfire CSV
27,828 records

Phase 1 — Data preprocessing

Load bands
Align SWIR 20m→10m

Compute indices
NDVI · NBR · dNBR

Severity mask
USGS 4-class

Build dataset
19.27M rows · Parquet

Feature filter
8 leak-free features

Phase 2 — Machine learning (per-pixel, 8 spectral features)

Spatial split
col 64/16/20

Logistic Reg.
Baseline · F1 0.883

Random Forest
200 trees · F1 0.884

MLP ★ Best
128→64→32 · F1 0.985

5-fold CV on MLP: Mean F1 = 0.9956 ± 0.0003 · Saved as .joblib pipelines

Phase 3 — CNN patch-based classification (BurnCNN, 9×9 px patches)

Extract patches · 237k · 9×9 px

Spatial split · col 64/16/20

BurnCNN · 3 conv blocks + GAP

Train 12 epochs · Adam · StepLR · 6.6 min

Final result: MLP F1 = 0.985 (best) · CNN F1 = 0.949 · RF F1 = 0.884 · LR F1 = 0.883

Burn Severity Distribution

Unburned	Low Severity	Moderate	High Severity
1,410.2 km²	**99.0 km²**	**168.0 km²**	**249.8 km²**
73.2% of valid pixels	5.1% of valid pixels	8.7% of valid pixels	13.0% of valid pixels

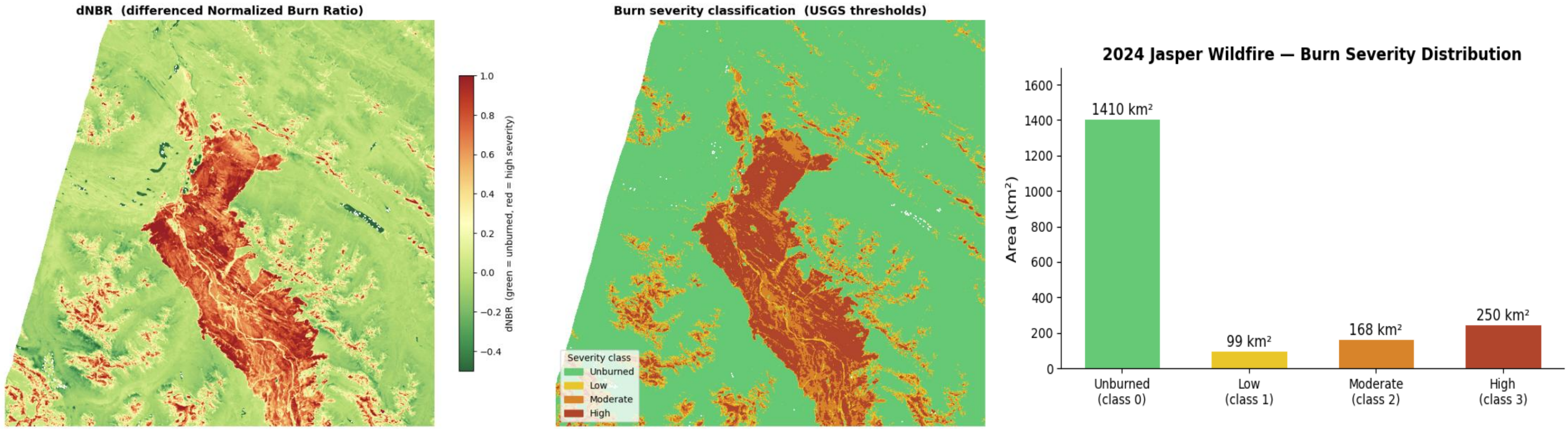
Total burned area: **516.7 km² (51,674 ha)** Provincial CSV records: 13,087 ha (park-managed areas not in provincial system)

Top feature correlations with burned label:

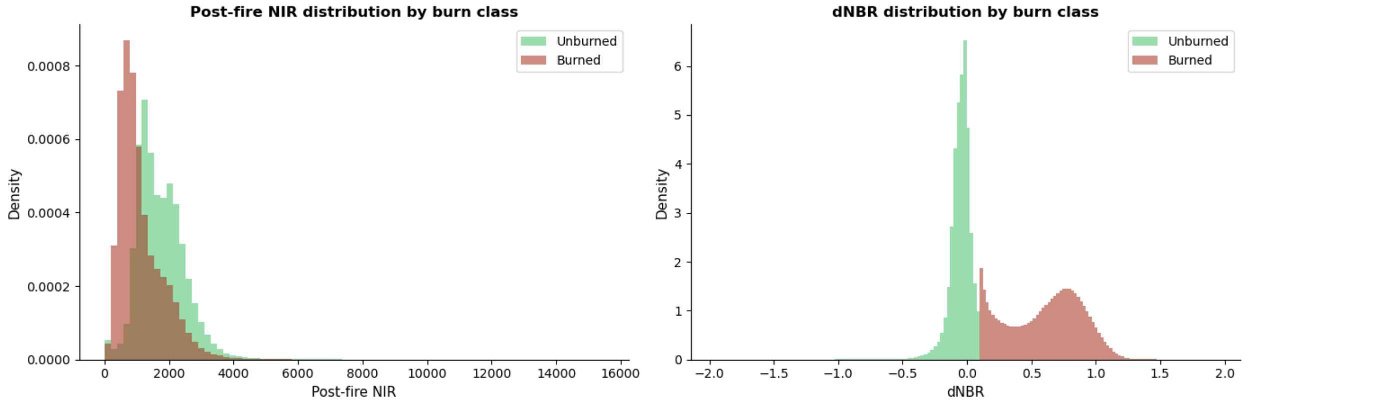
- **dNBR** r = 0.86 — EXCLUDED — IS the label threshold
- **ndvi_diff** r = 0.57 — EXCLUDED — directly encodes label
- **post_nir** r = 0.35 — INCLUDED — raw band
- **post_nbr** r = 0.64 — Excluded (conservative — NBR ratio)
- **post_ndvi** r = 0.52 — INCLUDED — single-date index
- **post_swir** r = 0.31 — INCLUDED — raw band

Key, C.H. & Benson, N.C. (2006). USGS burn severity classification thresholds.

2024 Jasper Wildfire — Burn Severity Assessment



Spectral separation between burned and unburned pixels



EXPERIMENTAL DESIGN

Spatial Split & Leak-Free Features

Spatial column split — no test pixel is adjacent to a training pixel

TRAIN 64%	VAL 16%	TEST 20%
cols 0 → 2,915	cols 2,916 → 3,700	cols 3,701 → 4,775

8 Leak-free features (all models)

pre_nir	pre_swir
pre_red	post_nir
post_swir	post_red
pre_ndvi	post_ndvi

Excluded — data leakage

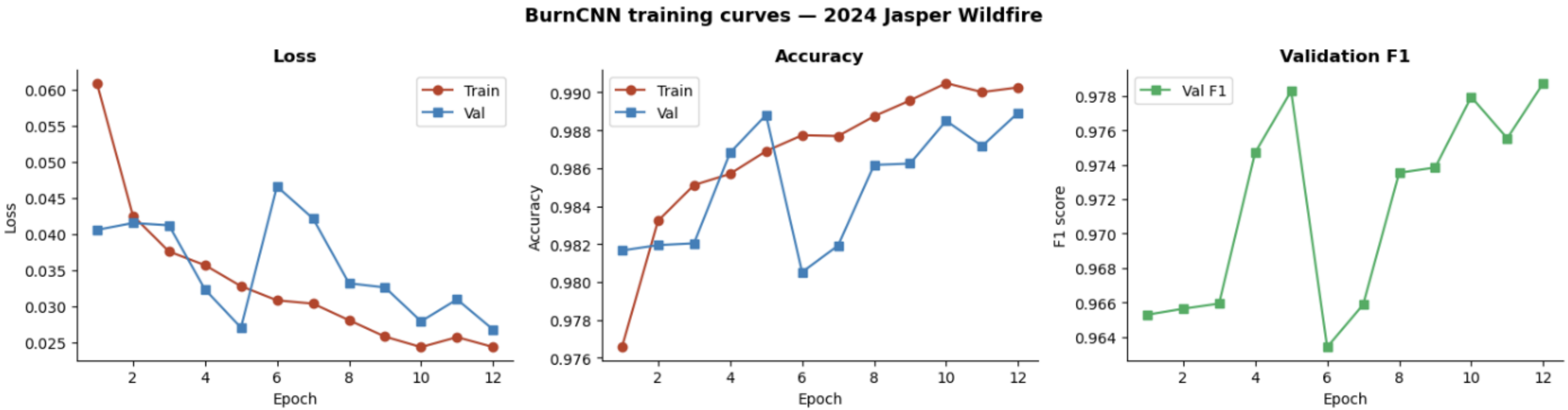
- ✗ dnbr — IS the label threshold (dNBR ≥ 0.10 = burned)
- ✗ ndvi_diff — directly encodes label direction
- ✗ pre_nbr, post_nbr — conservative removal

Split statistics

Train: 120,022 pixels — burned 56.6%

Val: 39,979 pixels — burned 58.1%

Test: 39,999 pixels — burned 22.0% (realistic imbalance)

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Training BurnCNN  —  12 epochs  |  lr=0.001  |  batch=256  |  device=cpu

 Epoch   Tr Loss   Tr Acc   Va Loss   Va Acc   Va F1   Time
──────────────────────────────────────────────────────────────
    1     0.0609   0.9766    0.0406   0.9817  0.9653    34s
    2     0.0425   0.9832    0.0415   0.9819  0.9656    32s
    3     0.0376   0.9851    0.0412   0.9820  0.9659    34s
    4     0.0357   0.9857    0.0323   0.9868  0.9747    32s
    5     0.0328   0.9869    0.0270   0.9888  0.9783    33s
    6     0.0308   0.9877    0.0466   0.9805  0.9634    36s
    7     0.0303   0.9877    0.0421   0.9819  0.9659    35s
    8     0.0281   0.9887    0.0332   0.9862  0.9735    34s
    9     0.0258   0.9896    0.0326   0.9862  0.9738    34s
   10     0.0243   0.9905    0.0279   0.9885  0.9779    35s
   11     0.0257   0.9900    0.0309   0.9872  0.9755    35s
   12     0.0243   0.9903    0.0268   0.9889  0.9787    35s

Total training time : 6.8 min
```

RESULTS

Four-Model Comparison — Test Set

Logistic Regression

Accuracy

97.83%

F1 Score

0.883

Kappa

0.871

Baseline

Random Forest

Accuracy

97.79%

F1 Score

0.884

Kappa

0.872

Ensemble

MLP Neural Network

Accuracy

99.73%

F1 Score

0.985

Kappa

0.984

Best Model ★

CNN (BurnCNN 9×9)

Accuracy

99.26%

F1 Score

0.949

Kappa

0.945

Patch-level

5-fold CV (MLP, train set): **Mean F1 = 0.9956 ± 0.0003** — stable, no overfitting

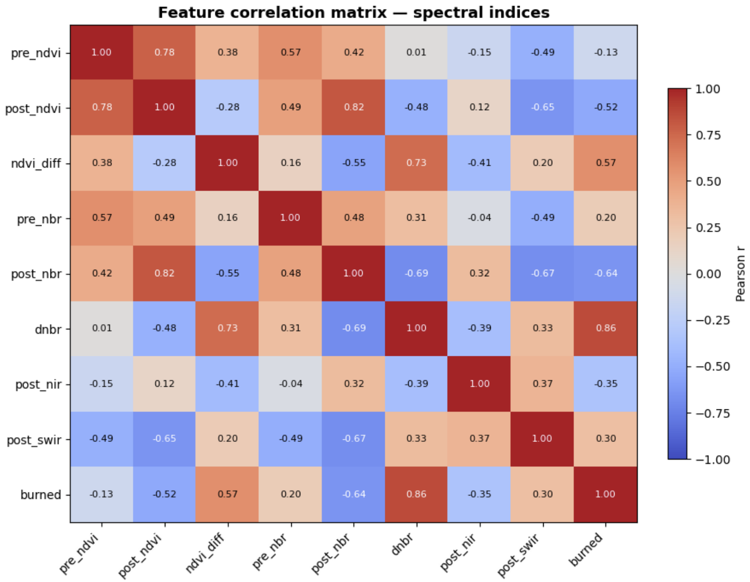
What Drives the Classification?

MLP — Permutation importance (mean F1 decrease)

post_swir	0.2428
post_nir	0.2288
pre_swir	0.1817
pre_nir	0.1407
post_ndvi	0.0285
pre_red	0.0282
post_red	0.0224

Random Forest — Mean decrease in impurity

post_ndvi	0.2516
post_nir	0.1384
post_swir	0.1319
pre_swir	0.1165
pre_ndvi	0.1124
pre_red	0.1022
post_red	0.0819



Key insight: MLP relies on post-fire SWIR/NIR raw reflectance; RF relies on post-fire NDVI. Both prioritise post-fire signal — confirming bi-temporal design is essential.

Key Findings and Study Limitations

Key Findings

1. 516.7 km^2 mapped vs. 13,087 ha in provincial CSV — difference explained by park-managed areas outside provincial jurisdiction
2. MLP achieves best per-pixel performance (F1 = 0.985, Kappa = 0.984) using only 8 raw spectral features
3. LR and RF are nearly identical (F1 ≈ 0.883) — spectral boundary is approximately linear for tree/linear models
4. CNN achieves F1 = 0.949 at patch level — spatial predictions are smoother and more suitable for mapping applications
5. CSV tabular features (temperature, humidity, wind) add no discriminating power when broadcast uniformly to all pixels

Limitations

Label construction

Labels derived from dNBR threshold on the same imagery used for training — future work should validate against independent fire perimeter shapefile

Single fire event

Analysis covers one fire complex; generalisability to other fuel types or sensors not tested

9×9 patch size (CNN)

Small receptive field limits spatial context; larger patches would improve CNN accuracy at higher CPU cost

CSV tabular fusion

Point-observation weather broadcast uniformly — spatially explicit raster weather data needed for meaningful fusion

CONCLUSION

What We Found

- Burned area mapping from raw bi-temporal Sentinel-2 bands alone achieves near-perfect accuracy when spatial leakage is properly controlled
- MLP Neural Network (F1 = 0.985) is the best model — capturing non-linear spectral relationships that LR and RF miss
- dNBR is the strongest predictor but must be excluded to prevent leakage — raw bands are sufficient
- CSV fire weather features provide environmental context and validation but cannot improve pixel-level spatial classification

99.73%

MLP accuracy

0.985

MLP F1 score

516.7

km² mapped